EXPERIMENT 3

POLYALPHABETIC CIPHER (VIGENÈRE CIPHER) – PYTHON

AIM

To implement the Vigenère (Polyalphabetic) Cipher substitution technique using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Read the keyword from the user.
3. Repeat the keyword until its length matches the plaintext.
4. Encrypt each character by adding the positions of the plaintext character and the corresponding key character modulo 26.
5. Display the encrypted (cipher) text.
6. Decrypt the cipher text by subtracting the key character value from the cipher character modulo 26.
7. Display the decrypted text.

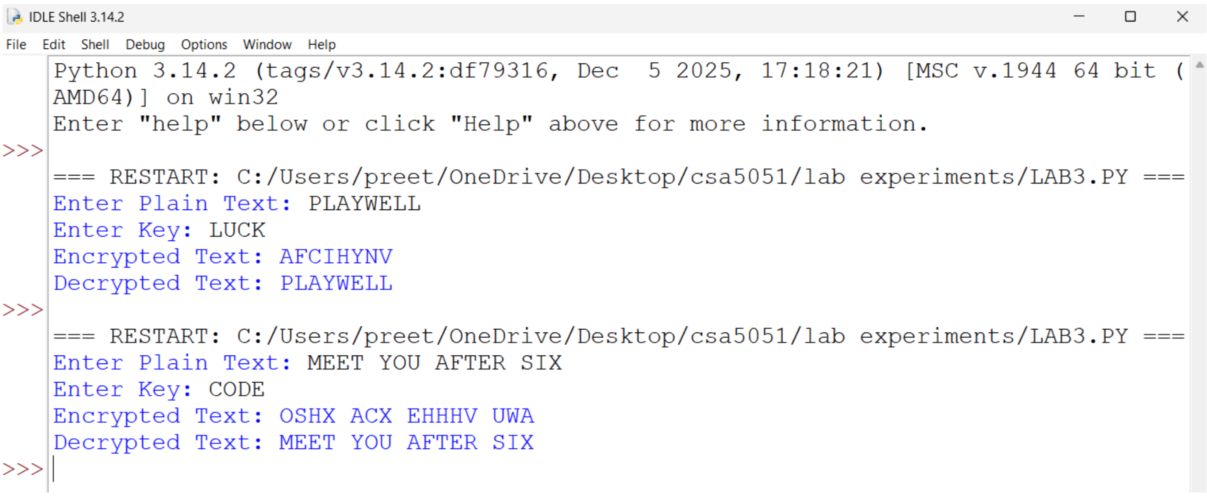
PYTHON CODE

```
def generate_key(text, key):
    key = key.upper()
    text = text.upper()
    new_key = ""
    j = 0

    for ch in text:
        if ch.isalpha():
            new_key += key[j % len(key)]
            j += 1
        else:
            new_key += ch
    return new_key
```

```python
def encrypt(text, key):
    text = text.upper()
    key = generate_key(text, key)
    cipher = ""

    for i in range(len(text)):
        if text[i].isalpha():
            x = (ord(text[i]) - ord('A') + ord(key[i]) - ord('A')) % 26
            cipher += chr(x + ord('A'))
        else:
            cipher += text[i]

    return cipher


def decrypt(cipher, key):
    cipher = cipher.upper()
    key = generate_key(cipher, key)
    plain = ""

    for i in range(len(cipher)):
        if cipher[i].isalpha():
            x = (ord(cipher[i]) - ord(key[i])) % 26
            plain += chr(x + ord('A'))
        else:
            plain += cipher[i]
```

```
    return plain



plaintext = input("Enter Plain Text: ")

key = input("Enter Key: ")


cipher = encrypt(plaintext, key)

print("Encrypted Text:", cipher)


plain = decrypt(cipher, key)

print("Decrypted Text:", plain)
```

OUTPUT



```
IDLE Shell 3.14.2
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB3.PY ===
Enter Plain Text: PLAYWELL
Enter Key: LUCK
Encrypted Text: AFCIHYNV
Decrypted Text: PLAYWELL
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB3.PY ===
Enter Plain Text: MEET YOU AFTER SIX
Enter Key: CODE
Encrypted Text: OSHX ACX EHHHV UWA
Decrypted Text: MEET YOU AFTER SIX
>>>
```

RESULT

Thus, the Vigenère (Polyalphabetic) Cipher substitution technique was implemented successfully using Python.